Journal of the Economic Science Association (2023) 9:157–175
https://doi.org/10.1007/s40881-023-00132-8

ORIGINAL PAPER

Demographic differences in the effect of price on giving in a diverse population

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Received: 17 February 2021 / Revised: 25 October 2022 / Accepted: 13 April 2023 /
Published online: 8 May 2023
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Abstract
We conduct two experiments using a demographically diverse online subject pool to investigate total and extensive price elasticities of giving by age, income, gender, political ideology, and religiosity. A first exploratory experiment finds that religious subjects give more, are more likely to give, and are less sensitive to the price of giving than non-religious subjects. We find no statistically significant differences in price elasticities by age, income, gender, or political ideology. A second pre-registered experiment confirms these findings.

Keywords Altruism · Price elasticity · Demographics · Dictator game · Mechanical Turk

JEL Classification C90 · C91 · C99 · D31 · D64

1 Introduction

Examining differences in behavior between demographic groups requires subject population heterogeneity in the demographic characteristic under study. Economic experiments conducted in a laboratory setting are typically located on college campuses and draw subjects from a student body that is heterogeneous in gender, which makes the studies of differences by gender practical. But by contrast, college students are largely homogeneous in age and income. Examining differences by non-heterogeneous characteristics in such a setting is at best difficult and at worst impossible. This leads to the research question addressed in this paper: Are there demographic differences in the level or price elasticity of giving in a demographically diverse subject pool?

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A substantial body of literature has examined gender differences in giving using both student and non-student subjects in the context of dictator games.[1,2] Brañas-Garza, et al. (2018) using subjects recruited through Amazon's mechanical Turk, Eckel and Grossman (1998) using undergraduate subjects, and Croson and Gneezy (2009) in a review of studies mostly using undergraduates, all found that women send more to an anonymous partner than do men. In contrast, Bolton and Katok (1995) and Cadsby, et al. (2010) using undergraduate subjects, Ben-Ner, et al. (2004) and Eckel, et al. (2011) using high school student subjects, and Henrich et al. (2005) examining behavior of adults in 15 societies, found no significant difference in sent amount between women and men. Andreoni and Vesterlund (2001) found that among undergraduate students women sent significantly more than men when the cost of giving was equal to or greater than 1 while men sent more when the cost of giving was less than 1.[3] In studies examining demographic differences beyond gender, Wang and Graddy (2008) using a large scale telephone survey found that subjects who said religion was an important part of their lives gave more to religious causes than did subjects who said religion was not an important part of their lives. However, they found no difference between the two groups in giving to secular causes. In a survey of adults in the Netherlands, Wiepking and Maas (2009) found that membership in a religious network increased the level of charitable giving. Eckel and Grossman (2004) found no significant difference in the level or price elasticity of giving between religious and non-religious subjects among undergraduate students. However, they found that religious subjects had a significantly higher income elasticity of giving than did the non-religious, suggestive of behavior similar to tithing.[4]

Other studies have analyzed how giving is affected by altering the effective price of giving through the use of matching funds or rebates.[5] In an online experiment using Mechanical Turk subjects, Gandullia and Lezzi (2018) found charitable donations were higher in a match treatment than in a mathematically equivalent rebate treatment. They also found that women in the match treatment gave more than men and Democrats gave more than Republicans, while subjects over age 55 gave more in both matching and rebate treatments. In a field experiment of charitable giving, Karlan and List (2007) found that providing matching funds increased the total amount donated, but higher match rates did not cause significant additional giving.

[1] In considering these studies and in our own we consider the terms "giving", "sent", and "send" to be synonymous.

[2] See Niederle (2015), Engel (2011), and Rand et al. (2016) for additional reviews of studies examining differences in giving behavior by gender.

[3] Cost defined as the ratio of the reduction of payoff to self divided by the increase in payoff to the other subject.

[4] The fundraising literature generally finds that people active in religious groups give more than those who are not. See Flanagan (1991).

[5] The difference between match and rebate can be clarified by the following examples where the subject chooses to send $1 to a recipient. In a one-to-one match the experimenter would add ("match") the same amount so the recipient would receive $2. In a rebate treatment, the recipient would receive the $1 sent by the subject, but then the experimenter would give $0.50 back to the subject as a "rebate". The two are mathematically equivalent in that the final cost to the subject of increasing the recipient's payoff by x is 0.5x.

They also report a large difference in response to the offer of a matching contribution based on the political orientation of the state of residence of the donor. Offering matching funds increased donation amounts in red states (states won by George Bush in 2004) but had no effect in blue states (states won by John Kerry in 2004).

We extend this literature by reporting the results of two experiments using a modified dictator game and a demographically heterogeneous online subject pool. Our first experiment is exploratory, estimating the level of giving, probability of giving, price elasticity of giving, (referred to here as *total price elasticity*), and the price elasticity of giving a non-zero amount, (referred to here as *extensive price elasticity*) by age, income, gender, political ideology, and religiosity. Our exploratory results lead us to four hypotheses; that the level of giving, probability of giving, total price elasticity, and extensive price elasticity of giving differ by subjects' religiosity. We conduct a second, pre-registered experiment to test these hypotheses using subjects with very high or very low levels of self reported religiosity. We find statistically significant evidence in support of the first three hypotheses, but not for a difference in extensive price elasticity. In neither study do we find significant differences with respect to other demographic variables.

The remainder of this paper is organized as follows. The next section presents the exploratory experiment. Section 3 presents the pre-registered experiment. Section 4 is discussion, and Sect. 5 concludes.

2 Exploratory experiment

2.1 Methods

The experiment was conducted online over a two week period in July of 2018 using Mechanical Turk workers in the United States as subjects.[6,7] After reading an IRB approved informed consent disclosure, subjects were presented with a task structured as a dictator game with a multiplication factor which increased or decreased the amount received by a randomly paired passive counterpart for any given amount sent. The multiplication factor μ for each subject was randomly drawn from a continuous uniform distribution $\mu \in (0, 2)$ shown to subjects as an integer percentage between 0 and 200% inclusive.[8] Subjects received an endowment of $1.00 and were instructed that they may send any portion of the endowment to a randomly selected subject in another group and that any amount they chose to send would be modified by the experimentor by the multiplication factor.[9] Subjects were shown the multiplier value which had been randomly assigned to them, and they were then shown examples of the amounts that would be received by the passive player if they sent $1.00, $0.80, $0.60, $0.40, $0.20, and $0.00. Subjects then entered the amount they

[6] Instructions and supplementary materials are available in the online appendix at https://www.opencpsr.org/openicpsr/project/181921/version/V1/view.

[7] Studies comparing behavior of online and in-person subjects typically find no significant difference (Buso et al., 2021; Li et al., 2021).

[8] A multiplier of 0% was shown to a single subject who then elected to send $0.

[9] Amir, Rand, and Gal (2012) and Raihani, Mace, and Lamba (2013) examined the effect on subject behavior of the lower stakes common in online experiments relative to the higher stakes common to in-person laboratory experiments. Both found no significant difference among U.S. subjects.

wished to send to the passive player. The next page asked for the subject's gender, age range, and income range. This concluded the main experiment.

Subjects completing the main experiment were subsequently contacted by email through Mechanical Turk inviting them to complete a Qualtrics survey to collect additional demographic data for use in a later, follow-on study and were paid a flat amount of $0.50.[10] However, the later study was discontinued and the data collected is included in our analysis here.[11] Interactions between dictators and recipients were done asynchronously with all subjects making their decisions on amounts to send first, then those amounts modified by the multipliers were sent to passive recipients a few days later. Passive recipients were not paid a show-up fee and were told their task was simply to "receive money" without further explanation.

2.2 Results

2.2.1 Sample demographics and attrition

922 Mechanical Turk workers accepted the task for the main experiment. In MTurk terminology this is called a *Human Intelligence Task* or *"HIT"*. Of the 922, 130 did not complete the main experiment. An additional 20 subjects gave incomplete or invalid answers to one or more questions yielding 772 valid observations. Combining subjects who were non-responsive for either reason gives a dropout rate of 16.3%. Of the 772 subjects completing the main experiment, 505 responded and completed the additional demographic survey for a response rate of 65.4%. Subjects completing the follow-up survey were older ($37.3, \sigma = 12.2$) than those who did not ($34.5, \sigma = 11.7$). All other demographic differences between subjects who completed the follow-up survey and those who did not were small and none were statistically significant at a confidence level of $\alpha = 0.05$.[12] There was not a statistically significant difference in the mean amounts sent by subjects who responded to the additional survey and subjects who did not. Mean amount sent by responders [non-responders] was $0.333 [$0.326] ($t(774) = -0.274, p = 0.780$).

Of the 505 complete cases, 248 subjects were female (49.1%) and 257 were male (50.9%). The median age group was 30—39 years and median income group was $40,000—$59,999. The sample was left of center politically (mean = 35.9, $\sigma = 29.3$ where 0 is "Very liberal" and 100 is "Very conservative") and more non-religious than religious (mean = 33.8, $\sigma = 36.0$, where 0 is "Not at all religious" and 100 is "Very religious"). To determine whether we were correct in our belief that a sample composed of college students would have been less diverse in age and income than a sample of MTurk workers, we asked the same demographic questions of 101 undergraduate business students. The results confirmed our initial assumption.[13,14]

[10] Mechanical Turk worker ID's were recorded and matched to make sure the same workers participated in both HITs.

[11] See Sect. 1.2 in the online appendix for demographic variables and coding.

[12] See Sect. 1.3.2 in the online appendix.

[13] See Sect. 1.4.5 in the online appendix.

[14] A full breakdown of students' demographic characteristics is shown in appendix tables A5–A11.

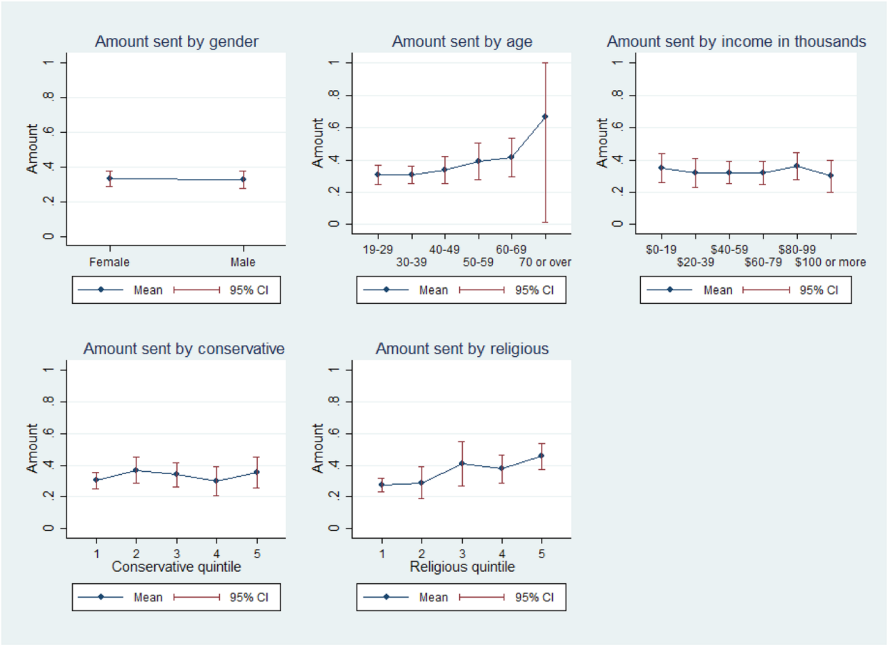

Fig. 1 Mean amounts sent by demographic groups

2.2.2 Correlation between variables

There were very high levels of correlation between different measures of political preference (Conservative, Party, Vote) and also between different measures of religiosity (Religious, Services).[15] To reduce multicollinearity, the continuous variables Conservative and Religious were retained while categorical Party, Services, and Vote were dropped. Variance inflation factors were calculated for the remaining variables in an OLS regression and none were above 1.2 which indicates low levels of multicollinearity.[16]

2.2.3 Level of giving

Subjects sent a mean of $0.33. 199 (39.4%) sent nothing, 79 (15.6%) sent their entire endowment of $1.00, and 227 (45.0%) sent a positive amount but less than their entire endowment. The mean amount sent conditional on sending a positive amount was $0.55.

Mean amounts sent by demographic groups are shown in Fig. 1. Tests of correlations between amounts sent and demographic groups are shown in Table 1. Testing

[15] See Sect. 1.6.1 in the online appendix.
[16] See Sect. 1.6.2 in the online appendix.

Table 1 Tests of correlations between amounts sent and demographic groups

Demographic variable	Test statistic
Gender	$t(n = 505, df = 503)$ $t(n = 505, df = 503) = 0.061$ [0.952]
Age	$\rho = 0.099^{\dagger}$ [0.025]
Income	$\rho = -0.009^{\dagger}$ [0.834]
Conservative	$\rho = 0.053^{\dagger}$ [0.232]
Religious	$\rho = 0.238^{\dagger}$ [< 0.001]

p values in brackets

†Spearman's rank correlation coefficient

for differences in amounts sent by demographic groups requires multiple comparisons with the associated elevated risk of type I error. Using the Bonferroni correction for five comparisons, an α level of 0.05 corresponds to a p value threshold of 0.01. Using this standard, there is a statistically significant positive correlation between amount sent and religiosity, but not age, income, or political ideology. The difference between mean amounts sent by women ($0.332) and men ($0.334) was also not statistically significant.

As a robustness check, amounts sent by demographic groups were also estimated using the model shown in Eq. 1, producing similar results as shown in Table 2, model 1

$$y_i = \alpha_0 + \alpha_1 X_i + \epsilon_i \tag{1}$$

where subject index $i = 1 \ldots, 505$. y_i is amount sent by subject i and X_i is a vector of demographic characteristics of subject i.

2.2.4 Probability of giving

Figure 2 displays the proportions of subjects sending a positive amount by demographic group while Table 3 reports tests of correlations between demographic characteristics and sending a positive amount.[17] Also in this case a Bonferroni correction for multiple comparisons was applied such that an α level of 0.05 corresponds to a p value threshold of 0.01. Females were significantly more likely to send a positive amount than males (67.3% vs. 54.1%). There was also a significant correlation between sending a positive amount and religiosity, but not with any other demographic variables.

[17] Table 2 model 4 reports similar results as a robustness check.

Table 2 Exploratory experiment regression results of amounts sent and probability of sending a positive amount

	OLS	OLS	OLS	Probit	Probit
Variable	(1)	(2)	(3)	(4)	(5)
ln(Price)		−0.399			−0.239
		(0.300)			(0.286)
ln(Income)			0.050		
			(0.411)		
Female	−0.009	0.323	0.475	0.335***	0.337*
	(0.032)	(0.173)	(0.924)	(0.117)	(0.123)
Age	0.003	0.018	0.036	0.009*	0.013
	(0.001)	(0.007)	(0.034)	(0.005)	(0.005)
Income	0.000	−0.001		−0.001	−0.001
	(0.001)	(0.003)		(0.002)	(0.002)
Conservative	0.000	−0.003	−0.018	−0.003	−0.003
	(0.001)	(0.003)	(0.017)	(0.002)	(0.002)
Religious	0.002***	0.011***	0.017	0.010***	0.008***
	(0.000)	(0.003)	(0.014)	(0.002)	(0.002)
ln(Price) × Female		0.036			0.011
		(0.160)			(0.142)
ln(Price) × Age		−0.008			− 0.010
		(0.005)			(0.006)
ln(Price) × Income		−0.001			− 0.001
		(0.003)			(0.002)
ln(Price) × Conservative		−0.003			0.003
		(0.003)			(0.003)
ln(Price) × Religious		0.009***			0.008**
		(0.002)			(0.002)
ln(Income) × Female			−0.037		
			(0.273)		
ln(Income) × Age			−0.006		
			(0.009)		
ln(Income) × Conservative			0.004		
			(0.004)		
ln(Income) × Religious			−0.001		
			(0.004)		
Constant	0.175**	−3.246***	−3.543	−0.412	−0.405
	(0.061)	(0.330)	(1.567)	(0.228)	(0.240)
R^2	0.060	0.126	0.067	0.072	0.120[a]
Observations	505	505	505	505	505

Model 1 regresses amount sent on demographic variables

Model 2 adds independent variables for log of price and demographic interactions

Model 3 regresses amount sent on log of income and demographic variables

Model 4 regresses sending a positive amount on demographic variables

Model 5 regresses sending a positive amount on log of price, demographic variables, and interaction terms

Table 2 (continued)

Standard errors in parentheses

*, ** and *** indicate 10, 5, and 1 percent significance levels after Bonferroni correction for multiple comparisons

[a]McFadden's pseudo R^2

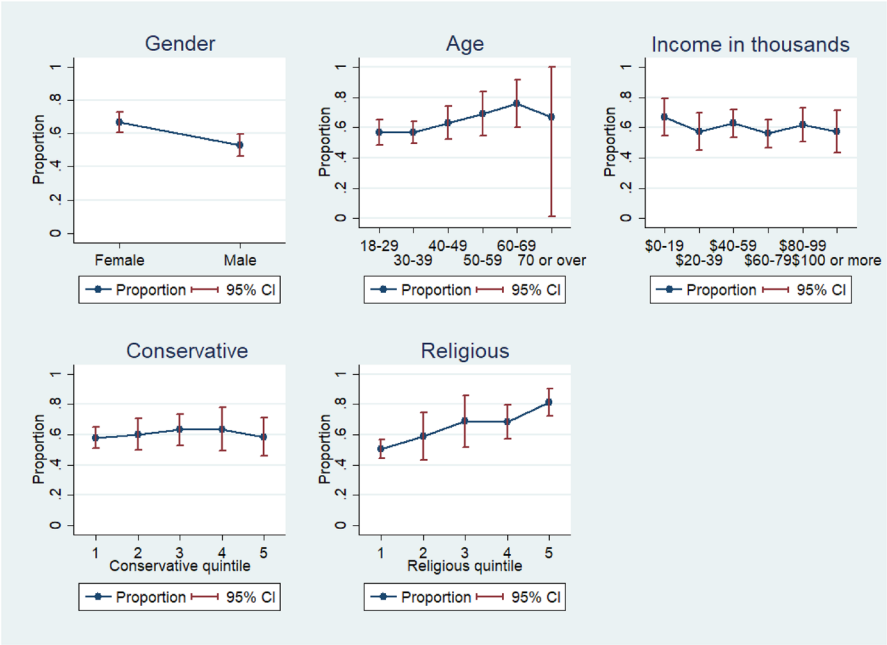

Fig. 2 Proportion sending a positive amount by demographic groups

Table 3 Tests of correlations between sending a positive amount and demographic groups

Demographic variable	Test statistic
Gender	$t(503) = -3.069$ [0.002]
Age	$\rho = 0.080^{\dagger}$ [0.073]
Income	$\rho = -0.016^{\dagger}$ [0.723]
Conservative	$\rho = 0.047^{\dagger}$ [0.288]
Religious	$\rho = 0.259^{\dagger}$ [< 0.001]

p values in brackets

[†]Spearman's rank correlation coefficient

2.2.5 Total price elasticity

Graphs of relationships between amounts sent and the multiplier for demographic sub-groups are shown in Fig. 3.

Total price elasticities η_t were estimated using the model shown in Eq. 2. Amounts sent and prices were expressed as logs for ease of interpretation as elasticities.

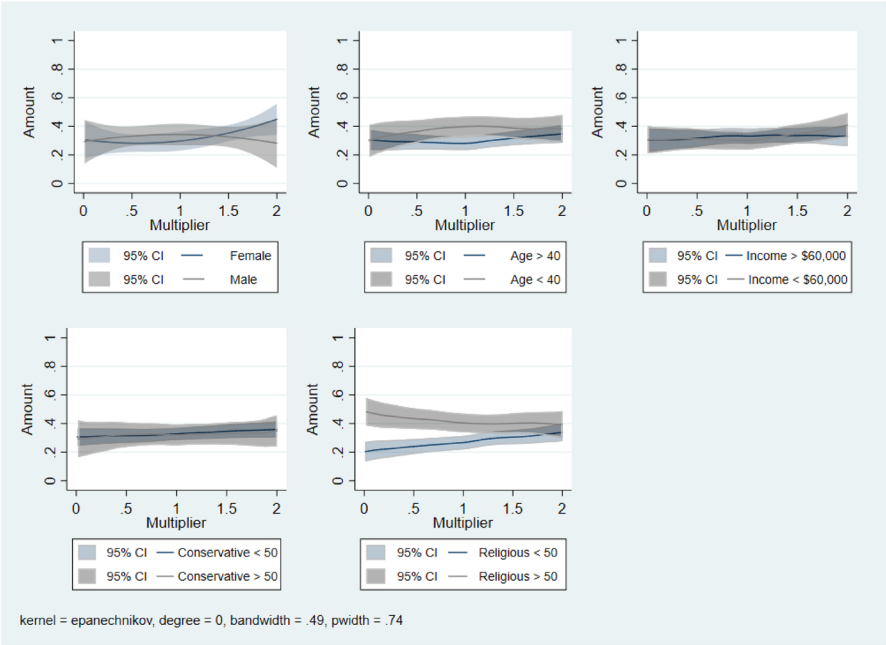

Fig. 3 Amount sent vs. multiplier for demographic sub-groups

$$lnY_i = \alpha_0 + \alpha_1 lnP_i + \alpha_2 X_i + \alpha_3 lnP_i \times X_i + \epsilon_i \tag{2}$$

where subject index $i = 1\ldots,505$. Y_i is amount sent by subject i plus one cent to avoid taking the log of zero, P_i is price for subject i calculated as the inverse of the multiplier thus expressing the cost to the dictator of having \$1 be received by the passive player. X_i is a vector of demographic characteristics of subject i. Total price elasticity regression results are shown in Table 2, model 2. For consistency with our analysis of the level of giving, we again apply the Bonferroni correction for multiple comparisons to the regression results.

Coefficient estimate of *Religious* is positive and statistically significant confirming that level of giving is increasing in religiosity. Coefficient estimates of the interaction terms between *ln(Price)* and *Female*, *Age*, *Income*, and *Conservative* show no statistically significant difference in total price elasticity in those demographic variables. The interaction between *ln(Price)* and *Religious*, however, indicates a highly statistically significant difference in total price elasticity by subjects' religiosity.[18]

Estimates of total price elasticities at representative demographic values are presented in Table 4, column 1 and Fig. 4. Total price elasticity with all variables held at their means is of the expected sign ($\eta_t = -0.507$). Estimates of total price

[18] As a robustness check, the analysis was repeated using a Tobit model with similar results presented in Sect. 1.7 of the online appendix.

Table 4 Estimates of elasticities at representative demographic values

Variable	Total price elasticity (1)	Extensive price elasticity (2)
All	−0.507**	−0.069**
	(0.228)	(0.030)
Male	−0.671**	−0.100**
	(0.341)	(0.045)
Female	−0.346	−0.038
	(0.296)	(0.037)
Age 25	−0.297	−0.036
	(0.310)	(0.041)
Age 65	−0.983*	−0.125**
	(0.531)	(0.062)
Income $20,000	−0.388	−0.055
	(0.349)	(0.044)
Income $100,000	−0.649*	−0.086*
	(0.381)	(0.049)
Conservative = 0	−0.547	−0.097**
	(0.357)	(0.044)
Conservative = 100	−0.437	−0.014
	(0.690)	(0.092)
Religious = 0	−1.367***	−0.165***
	(0.317)	(0.040)
Religious = 100	1.282**	0.074*
	(0.513)	(0.045)

Elasticities are calculated as marginal effects at the means of Eq. 2 for total price elasticity and of Eq. 4 for extensive price elasticity

Estimates are generated with the indicated variables set to the values shown and all other variables at their means

Standard errors in parentheses

*, ** and *** indicate 10, 5, and 1 percent significance levels

elasticity for the least religious subjects ($\eta_t = -1.367$) and the most religious subjects ($\eta_t = 1.282$) are both greater than one in absolute value but have opposite signs. As a robustness check, we divided the full sample into two subsets: those reporting religiousness less than 50 (n = 292), and those reporting religiousness greater than or equal to 50 (n = 164) on a scale of 1–100. We calculated the correlation between price and amount sent separately for each group again finding a significant negative relationship between price and amount sent for less religious subjects ($\rho = -0.105, p = 0.061$) but not for more religious subjects ($\rho = 0.130, p = 0.077$).

2.2.6 Income elasticity

We estimate income elasticities of giving using the model shown in Eq. 3.

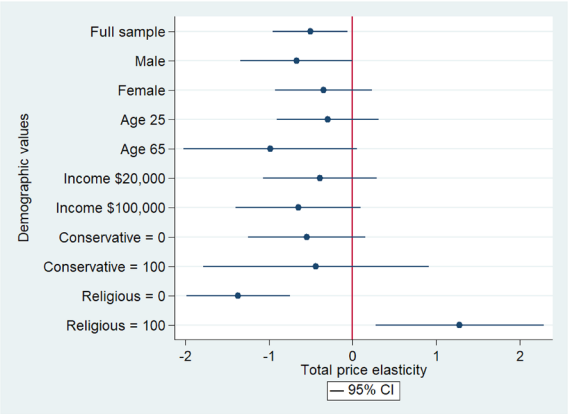

Fig. 4 Total price elasticities and confidence intervals for representative demographic values

$$lnY_i = \alpha_0 + \alpha_1 lnI_i + \alpha_2 X_i + \alpha_3 lnI_i \times X_i + \epsilon_i \tag{3}$$

where subject index $i = 1\ldots, 505$. Y_i is amount sent by subject i plus one cent to avoid taking the log of zero, I_i is income for subject I and X_i is a vector of demographic characteristics of subject i. Regression estimates are shown in Table 2 model 3. In contrast to Eckel and Grossman (2004), we find no statistically significant relationship between income elasticity and subject religiosity. We also find no significant difference in income elasticity for the sample as a whole or for any other demographic sub-group.

2.2.7 Extensive price elasticity

Analysis of subjects' decisions whether to send a positive amount is conducted with the semi-log probit model shown in Eq. 4.[19]

$$Pr\left(Y_i > 0\right) = \beta_0 + \beta_1 lnP_i + \beta_2 X_i + \beta_3 lnP_i \times X_i + \epsilon_i \tag{4}$$

Extensive price elasticity regression results are shown in Table 2, model 5. Coefficient estimates of the interaction terms between *ln(Price)* and *Female*, *Age*, *Income*, and *Conservative* show no statistically significant difference in extensive price elasticity in those demographic variables. The interaction term between *ln(Price)* and *Religious* shows a statistically significant difference in extensive price elasticity by subjects' religiosity. Estimates of extensive price elasticities at representative demographic values are presented in Table 4, column 2. Extensive price elasticity with all demographic variables held at their means is of the expected sign $\left(\eta_e = -0.064\right)$. As in the analysis of total price elasticity in the previous section, the estimates of

[19] See Eq. 2 for variable definitions.

extensive price elasticity for the least religious subjects ($\eta_e = -0.169$) and for the most religious subjects ($\eta_e = 0.078$) have opposite signs.[20]

3 Pre-registeres experiment

3.1 Method

To test the robustness of the main results of the exploratory experiment, we conducted a pre-registered experiment to determine if there is a statistically significant difference in the level, total price elasticity, and extensive price elasticity of giving between subjects with very high and very low degrees of self reported religiosity.[21]

The experiment was conducted over one week in the Spring of 2022 using Mechanical Turk workers in the US who had not participated in the exploratory experiment. After accepting the task, subjects were transferred to Qualtrics where they were presented with an attention test question which also served to filter out bot scripts.[22] Subjects failing the question were allowed to proceed but their work was rejected in MTurk and the responses were not used. Subjects were then asked the same five demographic questions used in the main analysis of the exploratory experiment; *gender, age, income, political ideology, and religiosity*. For subjects reporting intermediate levels of religiosity, defined as strictly between 5 and 95 on a 0–100 scale, the study concluded and they were paid the advertised HIT fee of $0.25.

Subjects reporting very high (≥ 95) or very low (≤ 5) levels of religiosity were allowed to proceed to a second stage where they were shown instructions for a dictator game with the amount sent modified by the experimenter by a multiplication factor. Subjects then viewed instructions which included a table of example amounts sent and received based on their assigned multiplier value.[23] The decision task differed from the task in the exploratory experiment in two important ways. First, after reading the instructions subjects were asked *"If your multiplier was* [treatment multiplier value]*% and you decided to send $1.00 to the other person, how much would they receive?"* as a test of instruction comprehension. Subjects failing the instruction comprehension test were allowed to proceed, but their responses were not included in our analysis. Second, instead of the multiplication factor being drawn from a continuous distribution, subjects were randomly assigned to one of two treatments; a low multiplication factor of 0.2, or a high multiplication factor of 2.0, thus creating a 2 by 2 design of high/low religiosity × high/low multiplier. The high multiplier of 2.0 implied a low price of $0.50 to have the second player receive $1.00, similarly the low multiplier implied a high price, but subjects were shown the treatment in terms of a multiplier percentage, not expressed as a price. After reading

[20] As in the previous section, the least and the most religious subjects are defined as *Religious* = 0 and *Religious* = 100.

[21] AEA RCT Registry, RCT ID: AEARCTR-0009459.

[22] See Sect. 1.1 in the online appendix.

[23] See Sect. 2.1 in the online appendix.

Table 5 Observations by religiosity and price treatment

	Price		
	High	Low	Total
Low religiosity	299	359	658
High religiosity	125	163	288
Total	424	522	946

instructions and answering the understanding question subjects decided how much of a $1 endowment they wished to send to an anonymous second player and the task concluded.

3.2 Results

3.2.1 Sample demographics and attrition

4483 MTurk workers accepted the HIT. Following the recommendation in Kennedy et al. (2020), we used iphub.info to screen IP addresses of respondents as recorded by Qualtrics against a list of known spam addresses and addresses concealing their physical locations through a virtual private network, virtual private server, or other methods to detect participants outside the US who had evaded MTurk's US resident filter. 223 observations failed this screen and were excluded from our analysis. An additional 83 responses were dropped where subjects failed to complete the experiment. 181 observations were dropped due to the subject failing the attention test/bot screen question leaving 3996 valid observations. Of the 3996 valid observations, 1229 (30.8%) reported religiosity either ≥ 95 (490, 12.3%) or ≤ 5 (739, 18.5%) and proceeded to the decision task stage. Of the 1229, 946 (77.0%) answered the understanding question correctly consisting of 424 of 611 (69.4%) in the 0.2 multiplier (high price) treatment and 522 of 618 (84.5%) in the 2.0 multiplier (low price) treatment. Answering the understanding question correctly was not independent of multiplier value ($\chi^2(1) = 39.38, p < 0.001$). Observations where subjects answered the understanding question incorrectly were excluded from further analysis. The final number of observations by religiosity and price are shown in Table 5.

Demographics of the 946 subjects were similar to those in the exploratory experiment with rough gender parity, 439 (46.4%) reporting female, 495 (52.3%) reporting male. 12 subjects (1.3%) reported "Other/Prefer not to state". The median age group was 30–39 and median income group was $40,000–$59,999. Subjects were left of center politically (meanonascaleof $1 - 100 = 33.6, \sigma = 31.9$) and predominantly non-religious (meanonascaleof $1 - 100 = 30.5, \sigma = 45.4$).[24]

[24] Full demographic data is shown in Sect. 2.2 in the online appendix.

3.2.2 Level of giving

The mean amount sent overall was \$0.43. Mean sent was \$0.50 in the low price treatment and \$0.35 in the high price treatment. The difference was statistically significant ($t(898) = 5.728, p < 0.0001$). Religious subjects sent significantly more than did non-religious subjects overall (\$0.60 *vs.* \$0.35, $t(550) = -8.979, p < 0.0001$); at the high price (\$0.50 *vs.* \$0.28, $t(222) = -5.130, p < 0.0001$), and at the low price (\$0.68 *vs.* \$0.41, $t(335) = -7.625, p < 0.0001$).

3.2.2.1 Result 1 *The exploratory experiment finding that religious subjects send more than non-religious subjects is confirmed.*

3.2.3 Probability of Giving

Religious subjects were also more likely to send a positive amount than were non-religious subjects overall (86% *vs.* 59%, $z = -8.113, p < 0.0001$); at the high price (79% *vs.* 46%, $z = -6.309, p < 0.0001$), and at the low price (91% *vs.* 70%, $z = -5.264, p < 0.0001$).

3.2.3.1 Result 2 *The exploratory experiment finding that religious subjects are more likely to send a positive amount than are non-religious subjects is confirmed.*

3.2.4 Total price elasticity

Total price elasticities η_t were estimated using the model shown in Eq. 5. Amounts sent and prices were expressed as logs for ease of interpretation as elasticities.

$$\ln Y_i = \alpha_0 + \alpha_1 \ln P_i + \alpha_2 \text{Religious}_i + \alpha_3 \ln P_i \times \text{Religious}_i + \varepsilon_i \quad (5)$$

Where subject index $i = 1\ldots, 946$. Y_i is amount sent by subject i plus one cent to avoid taking the log of zero, P_i is price for subject i calculated as the inverse of the multiplier thus expressing the cost to the dictator of having \$1 be received by the passive player. $Religious_i$ is a dummy variable indicating high religiosity group for subject i. Total price elasticity regression results are shown in Table 6, model 1. The coefficient estimate of the *Religious* dummy was positive and statistically significant, reaffirming Result 1. The coefficient estimate of the interaction term between *ln(Price)* and the *Religious* dummy was positive and statistically significant indicating that religious subjects choices were less price elastic than those of non-religious subjects. As a robustness check, the regression was repeated with additional demographic controls, shown in Table 6 model 2. Using this model the coefficient estimates of the *Religious* dummy and the interaction term between *ln(Price)* and the *Religious* dummy remained positive although for the latter at the $\alpha = 0.10$ significance level instead of $\alpha = 0.05$.

3.2.4.1 Result 3 *The exploratory experiment finding that total price elasticity varies by subjects' religiosity is supported at the 10% confidence levl.*

Table 6 Pre-registered experiment regression results

Variable	OLS (1)	OLS (2)	Probit (3)	Probit (4)
ln(Price)	−0.407***	−0.356***	−0.269***	−0.210***
	(0.029)	(0.094)	(0.017)	(0.079)
Female		0.459		0.417***
		(0.258)		(0.162)
Age		0.004		0.004
		(0.007)		(0.006)
Income		0.000		0.000
		(0.001)		(0.000)
Conservative		−0.004**		−0.003**
		(0.001)		(0.001)
Religious	1.151***	1.156***	0.846***	0.865***
	(0.107)	(0.061)	(0.103)	(0.073)
ln(Price) × Religious	0.099**	0.095*	0.045	0.060
	(0.036)	(0.053)	(0.040)	(0.065)
ln(Price) × Female		0.064		−0.010
		(0.055)		(0.035)
ln(Price) × Age		−0.001		−0.002
		(0.002)		(0.001)
ln(Price) × Income		0.000		0.001
		(0.001)		(0.001)
ln(Price) × Conservative		−0.002		−0.002
		(0.001)		(0.001)
Constant	−2.163***	−2.397***	0.328***	0.110
	(0.079)	(0.461)	(0.056)	(0.339)
R^2	0.128	0.151	0.099[b]	0.125[b]
Observations	946	934[a]	946	934[a]

Standard errors in parentheses

*, ** and *** indicate 10, 5, and 1 percent significance

[a]12 observations not reporting gender excluded

[b]McFadden's pseudo R^2

3.2.5 Extensive price elasticity

Extensive price elasticities η_e were estimated using the semi-log model shown in Eq. 6.[25]

$$Pr\left(Y_i > 0\right) = \beta_0 + \beta_1 lnP_i + \beta_2 Religious_i + \beta_3 lnP_i \times Religious_i + \epsilon_i \tag{6}$$

[25] See Eq. 5 for variable definitions.

Extensive price elasticity regression results are shown in Table 6, model 3. The coefficient estimate of the *Religious* dummy is positive and statistically significant reaffirming Result 2 that highly religious subjects were more likely to send a positive amount than non-religious subjects. The coefficient estimate of the interaction term between *ln(Price)* and the *Religious* dummy is not statistically significant. As a robustness check the regression was repeated with additional demographic controls shown in Table 6 model 4 with similar results.[26]

3.2.5.1 Result 4 *The exploratory experiment finding that extensive price elasticity varies by subjects' religiosity is not confirmed.*

4 Discussion

Our exploratory experiment finds that level and probability of giving are increasing in subjects' religiosity, while the absolute value of price elasticity of both decisions is declining in subjects' religiosity. Although we apply Bonferroni corrections to compensate for the increased risk of type I error due to multiple comparisons, we consider these exploratory results to be preliminary. To test the robustness of the exploratory results, we conduct a second experiment with a pre-registered design. The pre-registered experiment confirms the exploratory findings that compared to non-religious subjects, religious subjects give more, are more likely to give, and are less price sensitive in level of giving. However, the pre-registered experiment does not find a significant difference in extensive price elasticity and does not confirm that exploratory result.

Our findings that level and probability of giving are increasing in religiosity is consistent with a large body of studies in the philanthropy literature (see Bekkers and Wiepking (2011) for an overview). Our results are not consistent with Eckel and Grossman (2004), differing in two ways. First, they did not find a significant relationship between level or price elasticity of giving by religiosity. Second, they found a significant difference in income elasticity by religiosity, which we did not. However, Eckel and Grossman (2004) used an undergraduate subject pool rather than the MTurk subject pool used in our study. The difference in subject pools makes direct comparisons of their results to ours less certain. Our exploratory experiment results found that women gave more than men which agrees with the finding of a difference in giving level by gender in Brañas-Garza, et al. (2018). Both studies used MTurk subjects which makes the comparison more appropriate, however even with Bonferroni correction we are reluctant to claim that we confirm their results since differences in giving behavior by gender was not the focus of the pre-registered experiment and therefore not included in the pre-registration plan.

[26] Coefficient estimates of *Female* and *Conservative* were statistically significant in model 4. The former is consistent with meta-analyses reported in Rand et al. (2016) and Engel (2011) however, hypothesis tests of those demographic characteristics were not part of the pre-registered plan and are therefore not claimed here.

What could explain our finding that religious and non-religious subjects behave differently when presented with the opportunity to give at various prices to an anonymous person? While motivation was not the object of our study, we can speculate on several possibilities. If religious subjects place higher value on others' welfare, it would explain why they give more and are more likely to give. It is not obvious, however, how this would explain the difference in price elasticity. Some faiths commend the act of giving (e.g. tithing in many Christian denominations, zakat in Islam). This explanation, analogous to "warm glow" (Andreoni, 1989), would be more consistent with the observed behavior. It is also possible that the reduction in the price elasticity of giving displayed by religious subjects may have been due to a belief that giving when it was efficient to do so (i.e. at a multiplier greater than one) could serve both utilitarian and faith-based goals, while giving when it was inefficient served only the latter and thus provided greater warm glow for a faith motivated subject. Determining whether this was the cause is an open question for future studies.

Finally, notice that our experiments have some limitations, which should be kept in mind when considering the results. First, our initial intent was to use the data collected in the follow-up survey in a later paper but funding and scheduling issues caused us to alter our plan and fold the data into this study. This resulted in significant subject attrition from survey non-response. Also, even though we did not find any statistically significant differences in amounts sent or demographics between subjects who responded to the survey and those who did not, we cannot be certain there were not differences in other unobserved characteristics. Our judgement was that the benefit of using the survey data outweighed the costs but that judgment is open to disagreement. Second, we did not include an instruction comprehension test in the exploratory experiment out of our concern for increased subject drop-out due to task length balanced against our belief that the task was relatively simple. The fact that 23% of subjects did not correctly answer the instruction comprehension question in the pre-registered experiment suggests that our belief was incorrect. Finally, our finding that correctly answering the comprehension question in the pre-registered experiment was not independent of the multiplier value is concerning. If it was due to a difference in the mathematical difficulty of multiplying $\$1 \times 200\%$ versus $\$1 \times 20\%$, then it should not have introduced bias in our analysis, but although plausible we cannot be certain that was the cause. Whether this played any role in our failure to confirm a difference in extensive price elasticity by religiosity is unclear.

5 Conclusion

In this paper, we use a dictator game modified to include variation in price of giving to examine how choices vary by age, income, gender, political ideology, and religiosity. To this aim, we conduct a first exploratory experiment, and a second preregistered study, using a dictator game modified to include variation in price of giving to examine choices by age, income, gender, political ideology, and religiosity. The

central result of our experiments is a large and statistically significant difference in both level of giving and price responsiveness between religious and non-religious subjects, with religious subjects being more likely to give, giving more, and exhibiting lower total price elasticity.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s40881-023-00132-8.

Funding This work was funded by Concordia College Humanities and Social Science Scholars

Data availability Data and code used in this study are available from the Inter-University Consortium of Political and Social Research at https://doi.org/10.3886/E181921V1.

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